

Certificate-Driven Mechanization of Wakker–Debreu–Koopmans Additive Representation in Lean 4

Draft formal-methods companion to the Classical Lottery in Action artifact
(JAR / CPP / ITP / FormaliSE submission target)

May 18, 2026

Abstract

We report on a Lean 4 / Mathlib formalization of Wakker’s (1989) additive representation theorem (Theorem IV.2.7) and the Debreu–Koopmans (1960) concavity transfer theorem, structured around a *certificate decomposition* architecture that exposes the deepest construction outputs as named **Prop**-valued hypotheses. The artifact is sorry-free at every internal interface; closed forms supply weak order, coordinate-preference base independence, essentiality detection, tradeoff consistency, restricted solvability, the per-step standard-sequence shift, and the Archimedean axiom, all from the additive-representation predicate alone (Phase 0 necessity layer). The construction direction is factored into five named certificates whose discharge is targeted milestone-by-milestone. We discharge two of the five auxiliary residuals attached to the Wakker uniqueness certificate (*haff*) directly from essentiality of one coordinate plus standard additive-representation infrastructure, leaving the deep Wakker hexagon profile-construction obligation as the single remaining open hypothesis on that certificate. We argue the certificate decomposition is the right submission unit for monograph-scale mechanizations of measurement theorems whose paper-side argument spans several chapters.

1 Introduction and venue positioning

This paper is the formal-methods companion to a behavioral decision-theory artifact (*Classical Lottery in Action*, Management Science 2026) whose Lean 4 supplementary material imports the Wakker / Debreu–Koopmans (W/DK) representation layer as named **Prop**-valued hypotheses rather than as fully discharged theorems. The companion artifact discussed here mechanizes the imported W/DK layer itself. Its target audience is the formal-methods community: Journal of Automated Reasoning, Certified Programs and Proofs, Interactive Theorem Proving, Formal Aspects of Computing.

The methodological thesis is that monograph-scale measurement theorems of mathematical economics — where the paper-side proof spans dozens of intermediate lemmas across a multi-chapter argument — are best mechanized through an *explicit certificate architecture*: every construction output of the proof becomes a named **Prop** with documented mathematical provenance, and every consumer step plugs into those certificates with no hidden hypotheses. The architecture trades a clean axiomatic boundary for the partial discharge of a giant theorem, but at every snapshot the artifact is honest about what is and is not mechanized.

The contribution of the present paper to that thesis is twofold:

1. a precise specification of the certificate architecture for Wakker IV.2.7 + Debreu–Koopmans concavity transfer in Lean 4, including the necessity layer (Phase 0), the entry-point bundles

(Certificates 1–5), and the discharge regression theorems showing that any future certificate proof plugs into the existing public consumer interfaces unchanged;

2. the first round of partial discharges, including (a) a complete necessity bundle for Wakker IV.2.7 from the additive-representation predicate, (b) a structural-axiom-only grid-utility entry point, and (c) a chained discharge of the auxiliary residuals attached to Certificate 3 (uniqueness, *haff*) reducing the M2 frontier to a single named hexagon-condition obligation in the real-coordinate special case.

We do not claim full discharge of any of the five construction certificates. The remaining open content is exactly the genuine multi-month theorem-proving work that the certificate architecture isolates. We argue that the architecture itself, plus the Phase-0 necessity layer and the M2 partial discharge, is a genuine formal-methods contribution suitable for venues whose review criteria emphasize formalization architecture, audit honesty, and reusable infrastructure.

2 Companion artifact baseline

The Lean 4 baseline consists of three files in the companion artifact’s `Classical Lottery/` directory:

- B1. `WakkerInfrastructure.lean`:** fully proved infrastructure for product preferences, coordinate preferences, essentiality, restricted solvability, tradeoff consistency, standard sequences, Archimedean boundedness, two-coordinate modifications, convex preferences, and concavity of additive sums. This is the W/DK domain layer.
- B2. `WakkerExistence.lean`:** structural-axioms-only grid-utility entry point, mirroring Wakker IV.2.7 Step 2 without the `Function.update` bookkeeping carried in the main paper’s WDK file.
- B3. `WakkerDebreuKoopmans.lean`:** sorry-free wrapper and consumer theorems for Wakker IV.2.7 and Debreu–Koopmans hard direction. The file makes the deep missing work explicit by requiring construction certificates such as `hConstruct`, `hglobal`, `haff`, `hConc`, and `hPairConc`. It also contains the certificate roadmap table `WakkerRoadmap.CertificateChecklist.explicitCertificateCh` that mirrors the milestone layout of this paper.

The baseline is intentionally honest: it already type-checks the theorem interfaces and proves the gluing/consumer steps, while refusing to hide the monograph-scale arguments behind uninspected assumptions. Every public theorem passes the axiom audit [`propext`, `Classical.choice`, `Quot.sound`]; no rogue axioms are introduced anywhere in the artifact.

3 Certificate architecture

The methodological thesis is that large representation theorems in decision theory can be mechanized cleanly by splitting them into *certificates*: local construction outputs with precise Lean types, explicit mathematical provenance, and independently checkable consumer theorems.

3.1 Why certificates instead of monolithic proofs

Wakker IV.2.7 and Debreu–Koopmans hard direction are each the conclusion of a chapter-length argument. A monolithic mechanization has two failure modes that a certificate decomposition avoids:

- **Opaque assumptions.** Without explicit certificate boundaries, intermediate lemmas accumulate hidden hypotheses (e.g. a non-trivial standard-sequence calibration that turns out to require Archimedean witnesses) that are difficult to audit at submission time.
- **Reviewer cost.** A monolithic proof of a multi-chapter theorem cannot be reviewed in isolation; the certificate decomposition lets a reviewer assess the architecture, the necessity-layer audits, and the consumer theorems independently, which matches how formal-methods venues actually review submissions.

For Wakker/Debreu–Koopmans, the certificate architecture has two further advantages:

- It separates the decision-theoretic content from routine product-space, convexity, and finite-sum algebra already handled by Mathlib and the existing Lean infrastructure.
- It yields a sequence of intermediate, defensible formal results rather than a single opaque attempt to mechanize Wakker IV.2.7 in one pass. Each certificate discharge is itself a publishable milestone.

3.2 The Phase-0 necessity layer (proved)

Before attacking the construction direction, the artifact closes a complete *necessity* layer in `WakkerInfrastructure` and a structural-axioms-only grid-utility entry point in `WakkerExistence.lean` §1. Both files build clean against the same Mathlib pin as the main classical-lottery submission.

Necessity bundle (`WakkerInfrastructure.lean` §11). Define `AdditivelyRepresents P V` to assert that profile preference is decided by the sum of coordinate utilities, and `WakkerConstructionCertificate P` to assert the existence of such a V . The following are then proved as Lean theorems:

- `isWeakOrder_of_additivelyRepresents`: any additively represented preference is a weak order.
- `coordPref_iff_of_additivelyRepresents`: $\succeq_j$ is decided by V_j alone.
- `coordPref_base_independent_of_additivelyRepresents`: $\succeq_j$ is independent of the ambient profile.
- `Vj_nonconstant_of_essential_of_additivelyRepresents` and `essential_of_Vj_nonconstant_of_additivelyRepresents`: essentiality of coordinate j is exactly non-constancy of V_j .
- `tradeoffConsistency_of_additivelyRepresents`: the hexagon condition (Wakker IV.2.5) is automatic from additive representation.
- `restrictedSolvability_of_additivelyRepresents`: structural restricted solvability follows from the analytic `CoordUtilitySolvability` hypothesis on every V_j .
- `standardSequence_Vj_step`: every standard sequence step shifts V_j by the constant $V(\sigma.k, \sigma.r) - V(\sigma.k, \sigma.s)$.

- `archimedean_of_additivelyRepresents`: the Archimedean axiom holds in every coordinate.
- `certificate_necessity_bundle`: bundles the four global conclusions (weak order, tradeoff consistency, Archimedean, restricted solvability) under the certificate plus the analytic-solvability hypothesis.

Grid-utility entry point (`WakkerExistence.lean §1`). The reverse-direction seed lives in a separate file so that the artifact is independently buildable. It mirrors Wakker IV.2.7 Step 2:

- `coord_utility_on_grid_from_axioms`: from any injective standard sequence σ on coordinate j , build $V : X_j \rightarrow \mathbb{R}$ with $V(\sigma_\alpha n) = n$.
- `grid_utility_zero`: refined form anchoring $V(\sigma_\alpha 0) = 0$ and $V(\sigma_\alpha(n+1)) = V(\sigma_\alpha n) + 1$.
- `grid_utility_strictMono`: strict monotonicity of V along the grid.

Phase-0 audit row. The bundle is mirrored in `WakkerRoadmap.CertificateChecklist.explicitCertificate` (`WakkerDebreuKoopmans.lean`) with status `consumerReady`. This makes the necessity layer visible in the same place as the open construction certificates.

What the necessity layer rules out. Because the necessity direction is fully closed, every future attempt to build the construction certificate from the seven structural axioms can plug into the necessity layer for free: the produced V is automatically forced to satisfy each downstream property, so partial proofs that only establish weak order or only Archimedean become unnecessary. This collapses Wakker IV.6 into the artifact’s Phase-0 deliverable and lets later phases focus on Steps 1–5 of Wakker IV.2.

4 The five construction certificates

The construction direction is organized around five named certificates. Each one replaces an explicit hypothesis currently consumed by the Lean wrappers.

Certificate 1 (Wakker construction certificate). *Build coordinate utility functions from standard sequences and prove the pairwise slice representation currently supplied to `pairwise_additivity`. This is the formal core of Wakker’s tradeoff-measurement construction. Hypothesis name: `hConstruct`.*

Certificate 2 (Wakker global-gluing certificate). *Derive the global representation certificate `hglobal` from compatible pairwise representations and the dimensional hypothesis $\#i \geq 3$. This certificate discharges the hypothesis used by `global_additive_from_pairwise` and `wakker_IV_2_7_consumer`.*

Certificate 3 (Wakker uniqueness certificate). *Derive the affine-equivalence certificate `haff`: any two additive representations of an essential-coordinate product preference agree up to common positive scale and coordinate-specific translations.*

Certificate 4 (Debreu–Koopmans two-coordinate certificate). *Upgrade the currently available quasi-concavity and convex-upper-contour facts to true component concavity on two-coordinate slices under the required continuity and global-structure hypotheses. Hypothesis name: `hConc`.*

Certificate 5 (Debreu–Koopmans transfer certificate). *Derive `hPairConc` / `hConcAll` from convex preference plus additive representation, replacing the direct concavity certificate consumed by `concave_transfers` and `debreu_koopmans_hard_consumer`.*

5 Partial discharge of Certificate 3 (uniqueness, haff)

The first focused milestone of the construction direction is the discharge of Certificate 3, the Wakker uniqueness certificate. This section reports the current state.

5.1 The reference state

The public consumer `additive_rep_unique` consumes `haff` directly. The certificate predicate is (with $\iota \rightarrow i$, $\mathbb{R} \rightarrow \mathbb{R}$, etc., transliterated to ASCII for the verbatim block):

```
def AdditiveAffineUniquenessCertificate {X : i -> Type v}
  {P : ProductPref X} (R1 R2 : AdditiveRep P) : Prop :=
  exists (a : R) (_ : 0 < a) (b : i -> R),
  forall i x, R2.V i x = a * R1.V i x + b i.
```

The artifact factors this through the strictly weaker `AdditiveCommonScaleCertificate`:

```
def AdditiveCommonScaleCertificate {X : i -> Type v}
  {P : ProductPref X} (R1 R2 : AdditiveRep P) : Prop :=
  exists a : R, 0 < a /\
  forall (i : i) (u v : X i),
  R2.V i u - R2.V i v = a * (R1.V i u - R1.V i v).
```

The bridge `additiveAffineUniqueness_of_commonScale` discharges the uniqueness certificate from common-scale plus essentiality of all coordinates. The next layer factors common-scale through `TradeoffTransferCertificate`, which packages Wakker’s hexagon-condition cardinal-tradeoff equivalence:

```
def TradeoffTransferCertificate
  (R1 R2 : AdditiveRep P)
  (j : i) (v w : X j) (_hne : R1.V j v <> R1.V j w) : Prop :=
  forall (k : i) (u1 u2 : X k),
  R1.V k u1 <> R1.V k u2 ->
  (R2.V k u1 - R2.V k u2) / (R1.V k u1 - R1.V k u2) =
  (R2.V j v - R2.V j w) / (R1.V j v - R1.V j w).
```

The bridge `additiveCommonScaleCertificate_of_tradeoffTransfer_real` discharges the common-scale certificate (in the real-coordinate special case $X_i = \mathbb{R}$) from tradeoff-transfer plus three auxiliary inputs:

- **hne**: the reference pair (v, w) has distinct R_1 -values, witnessing essentiality of coordinate j under R_1 ;
- **hpos**: the cross-representation ratio at the reference pair is strictly positive;
- **hzero**: equal R_1 -values at any pair force equal R_2 -values (already discharged by `zero_difference_preservation`).

5.2 This paper’s M2 contribution

We discharge the first two of the three auxiliary inputs (**hne** and **hpos**) directly from essentiality of coordinate j , reducing the M2 frontier in the real-coordinate special case to a single named hypothesis: the tradeoff-transfer certificate itself.

The new theorem is

```

theorem additiveCommonScaleCertificate_of_tradeoffTransfer_real_from_essential
  {i : Type u} [Fintype i] [DecidableEq i]
  {P : ProductPref (fun _ : i => R)} (R1 R2 : AdditiveRep P)
  {j : i} (hess : ProductPref.Essential P j)
  (htransfer :
    forall {v w : R} (hne : R1.V j v <> R1.V j w),
      TradeoffTransferCertificate R1 R2 j v w hne) :
  AdditiveCommonScaleCertificate R1 R2.

```

The proof extracts the reference pair from the essentiality witness via the standard single-coordinate decomposition of `R.represents`, derives positivity of the cross-representation ratio via `positive_ratio_of_essential` and chains into the existing `additiveCommonScaleCertificate_of_tradeoffTransfer_real` bridge.

The companion theorem `additive_rep_unique_of_tradeoffTransfer_real_from_essential` threads this through `additiveAffineUniqueness_of_commonScale` and the public `additive_rep_unique` consumer, producing the full affine-equivalence conclusion of `additive_rep_unique` from

- essentiality of all coordinates,
- the existence of two essential coordinates (a hypothesis of the public consumer),
- essentiality of one chosen coordinate j , and
- the tradeoff-transfer certificate at j for arbitrary reference pairs (v, w) with $R_1.V_j v \neq R_1.V_j w$.

Both theorems pass the standard axiom audit [`propext`, `Classical.choice`, `Quot.sound`].

5.3 What remains open on Certificate 3

The single remaining open hypothesis on Certificate 3, in the real-coordinate special case, is the tradeoff-transfer certificate `TradeoffTransferCertificate R1 R2 j v w hne` for arbitrary reference pairs. We further factor this hypothesis through a corrected predicate.

Diagnostic obstruction (recorded honestly). An earlier attempt at this factoring introduced a `UtilityValueRealizingEquivalence_diagnostic` predicate that was provably unattainable from any nontrivial additive representation: `tradeoff_equivalence_difference_equality` forces the equivalence-pair difference to match the fixed reference-pair difference, while the realization clause forces it to match an arbitrary prescribed difference; the two together collapse the certificate to a trivially false statement. The in-file refutation `additiveBoolReal_not_utilityValueRealizingEquivalence` formalizes the obstruction at $\iota = \text{Bool}$, $X = \mathbb{R}$ with $(v, w) = (1, 0)$ on `false` and $(u_1, u_2) = (2, 0)$ on `true`; the meta-theorem `utilityValueRealizingEquivalence_diagnostic_unattainable_from_additivity` packages it as the standard “additively represented yet refutes the predicate” pattern.

Corrected predicate. The corrected `UtilityValueRealizingEquivalence` (no `_diagnostic` suffix) lets the reference pair (v', w') on j vary with the prescribed (u_1, u_2) on k , so the indifference-forced difference equality is satisfiable by choosing (v', w') with the matching $R_1.V_j$ -difference. Three named bridges constitute the corrected M2 frontier:

1. `OffDiagonalTradeoffTransferCertificate`: cross-coordinate ratio agreement at $k \neq j$.
2. `OnCoordinateRatioConsistency`: on-coordinate cardinal-equivalence statement at $k = j$.

3. **UtilityValueRealizingEquivalence** (corrected): existence of tradeoff equivalences with varied (v', w') and realized (u''_1, u''_2) endpoints.

The off-diagonal piece is fully discharged from the corrected predicate plus on-coordinate ratio consistency at (v, w) (the latter is an explicit companion hypothesis matching the cyclic structure of Wakker's proof). Theorem `offDiagonalTradeoffTransferCertificate_of_utilityValueRealizingEquivalence` mechanizes this.

The on-diagonal piece now has two theorem-backed discharge routes in the artifact. The first is the previous triangle argument: `onCoordinateRatioConsistency_of_triangle_through_third_coordinate` discharges it from two off-diagonal certificates chained through any third coordinate $m \neq j$. The second is a direct same-coordinate route: `CoordinateAffineLiftCertificate` isolates the stronger conclusion that on coordinate j itself one has an affine relation $R_2.V_j = \alpha R_1.V_j + \beta$ with $\alpha > 0$, and `onCoordinateRatioConsistency_of_coordinateAffineLift` then yields `OnCoordinateRatioConsistency` immediately. This is the cleaner target for the connected-coordinate standard-sequence calibration in Wakker's proof. The theorem `coordinateAffineLiftCertificate_of_commonStandardSequenceCalibration` proves this target from an already-normalized shared grid calibration, continuity of both utilities, and density of the grid; `coordinateAffineLiftCertificate_of_pairwiseGridNormalizationWitnesses` packages that normalized route through the existing `PairwiseGridNormalizationWitness` API. The stronger construction-side theorem `coordinateAffineLiftCertificate_of_strictStandardSequence` removes the over-normalization: for any strict standard sequence, `additiveRep_standardSequence_Vj_arithmetic` proves that each additive representation is an affine arithmetic progression on the same $\sigma.\alpha$ -grid, and `additiveRep_standardSequence_step_negative_of_strict` makes the step ratio positive. Continuity plus grid density then extend the affine relation from the grid to all of coordinate j . Theorems `coordinateAffineLiftContinuityInputs_of_monotone_rationalImage` and `standardSequenceGridDensity_real_of_betweenPointsCertificate` separately discharge those two inputs from the construction-facing certificates. The combined theorem `coordinateAffineLiftCertificate_of_strictStandardSequence` wires the direct route to `CoordinateMonotonicityCertificate`, `CoordinateRationalImageCertificate`, and `StandardSequenceGridBetweenPointsCertificate`; the sharper wrapper `coordinateAffineLiftCertificate_of_strictStandardSequence` derives those inputs from `SingleCoordinateMonotonicityAxiom`, rational-image coverage for the two representations, and `StandardSequenceGridDensityCertificate`. The older surjectivity-based wrapper remains available, but coordinate surjectivity is now itself theorem-backed from structural monotonicity plus rational-image coverage.

The end-to-end consumer `tradeoffTransferCertificate_of_utilityValueRealizingEquivalence_corrected` produces `TradeoffTransferCertificate` from the corrected predicate plus on-coordinate ratio consistency at (v, w) . A companion theorem `tradeoffTransferCertificate_of_utilityValueRealizingEquivalence_corrected_and_coordinateAffineLift` records the stronger direct route where the same-coordinate calibration on j is already available as `CoordinateAffineLiftCertificate`. The cyclic dependency between off-diagonal at (v, w) and on-coordinate ratio consistency at (v, w) therefore has two honest exits in the formal artifact: in the connected-coordinate case, direct standard-sequence calibration on j should produce the affine-lift certificate; in the discrete-coordinate case, the proof can still factor through a triangle.

Further factoring of the corrected predicate. The corrected `UtilityValueRealizingEquivalence` itself decomposes into two named sub-bridges plus the structural axiom `RestrictedSolvability`:

- **JDifferenceRealizationCertificate**: for any (k, u_1, u_2) with non-trivial $R_1.V_k$ -difference, find $(v', w' : X_j)$ with $R_1.V_j v' - R_1.V_j w' = R_1.V_k u_1 - R_1.V_k u_2$. This is the standard-sequence-density content of Wakker's proof on coordinate j .

- **TradeoffBracketingForallCertificate:** `TradeoffBracketingCertificate` quantified over all relevant reference pairs and primaries. This is the *Archimedean + monotone-grid* content already factored at the per-triple level in the artifact.

The discharge theorem `utilityValueRealizingEquivalence_corrected_of_jDifferenceRealization_and` produces the corrected predicate from these two sub-bridges plus `RestrictedSolvability`, by combining `tradeoffEquivalence_of_restrictedSolvability_and_bracketing` with the difference-realization step.

Discharge of `TradeoffBracketingForallCertificate` from coordinate-utility unboundedness. We further factor the bracketing-forall sub-bridge through the named hypothesis `CoordinateUtilityUnboundedness`. `R k`: the per-coord utility $R.V_k : X_k \rightarrow \mathbb{R}$ takes arbitrarily large positive values and arbitrarily large negative values. Under any additive representation, the bracket inequalities at the structural level (`P.weakPref`) reduce algebraically to $R.V_k u_{lo} \leq r \leq R.V_k u_{hi}$ for the prescribed real $r := R.V_k u_1 + R.V_j w' - R.V_j v'$, which the unboundedness certificate satisfies directly.

The discharge theorem `tradeoffBracketingForallCertificate_of_coordinateUtilityUnboundedness` composes a per-triple discharge `tradeoffBracketingCertificate_of_coordinateUtilityUnboundedness` over every coordinate. On real-coordinate domains, the stronger image-coverage certificates already present elsewhere in the artifact now feed this residual automatically: `coordinateUtilityUnboundedCertificate` derives unboundedness from interval-hitting of the image, `tradeoffBracketingForallCertificate_of_coordinateUtilityUnboundedness` packages that route into the full bracketing-forall certificate, and `tradeoffBracketingForallCertificate_of_coordinateUtilityUnboundedness` records the important corollary that the standard-sequence refinement target `CoordinateRationalImageCertificate` already suffices for the M2 bracketing side.

Stronger route via coordinate surjectivity. We also add a theorem-backed stronger real-coordinate route that closes both corrected M2 sub-residuals at once. If each coordinate utility is surjective onto $\mathbb{R}$ (certificate `CoordinateSurjectivityCertificate`), then `jDifferenceRealizationCertificate` realizes any prescribed difference on coordinate j by choosing one point with value $R_1.V_k u_1 - R_1.V_k u_2$ and one point with value 0. The companion theorem `tradeoffBracketingForallCertificate_of_coordinateSurjectivity` routes surjectivity through the already-proved interval-hitting coverage bridge to close the full bracketing side. Finally, `utilityValueRealizingEquivalence_corrected_of_coordinateSurjectivityCertificate` combines these two discharges with `RestrictedSolvability`, producing the corrected predicate directly from the stronger surjectivity hypothesis.

Sample witness. The canonical additive sum order on $\text{Bool} \rightarrow \mathbb{R}$ (formalized as `additiveBoolReal_pref` with the identity per-coord utilities `additiveBoolReal_rep`) satisfies the unboundedness certificate trivially: for every $r \in \mathbb{R}$, take $u_{lo} = r-1$ and $u_{hi} = r+1$. Theorem `additiveBoolReal_tradeoffBracketingForallCertificate` chains this sample through the discharge to validate that the predicate is non-vacuous.

The remaining open content on the corrected M2 frontier is now sharper still: common same-grid calibration, the continuity-input bridge, integer refinement, interval solvability, and the corrected refined mesh-family target are theorem-backed at the consumer/conditional level. `CoordinateMonotonicityCertificate` follows from the raw structural `SingleCoordinateMonotonicityAxiom`; `CoordinateUtilitySolvabilityCertificate` follows either from full-coordinate continuity by IVT or directly from `CoordinateSurjectivityCertificate`; and `CoordinateStandardSequenceIntegerRefinementCertificate` no longer needs coordinate surjectivity at this bridge layer: `CoordinateIntegerSeedCertificate`, supplied by integer- or rational-image coverage, plus `CoordinateStandardSequenceExtensionData` suffices. The theorem `coordinateStandardSequenceIntegerRefinementCertificate_of_rationalImage_and_extensionData` records the non-surjective route, and the integer-refinement certificate still implies two-sided integer

refinement and hence `CoordinateRationalImageCertificate` once interval solvability is available. Meanwhile `additiveRealBool_not_selectedRefinedDenseGridCertificate_true/false` shows that no single strict standard sequence can be the dense-grid target in the additive-real model: strict standard sequences there are arithmetic progressions. The replacement is `CoordinateUtilityRefinedMeshFamilyC`, a rational-indexed family of strict standard sequences whose utility images hit every real interval. It is theorem-backed from rational-image coverage plus raw one-step extension data and implies both between-points coverage and dense range. The raw Wakker seam is now split explicitly into two construction outputs: `CoordinateRationalRefinementBisectionCertificate`, whose standard-sequence rational hits imply rational-image and seed coverage, and `CoordinateConnectedContinuityOneStepBrac`, which together with `RestrictedSolvability` proves `CoordinateStandardSequenceExtensionData`. The theorem `rationalRefinementBisection_and_connectedContinuity_of_continuity_standardSequenceIn` proves both outputs from the current theorem-backed continuity plus calibrated integer-refinement machinery; conversely, `integerRefinement_and_fullContinuity_of_refinementBisection_connectedContinu` proves the two lower targets directly from rational refinement/bisection, connected-continuity one-step bracketing, restricted solvability, and structural monotonicity. The new reductions `coordinateConnectedCont`, `rationalRefinementBisection_and_connectedContinuity_of_rationalImage_and_extensionData`, and `rationalRefinementBisection_and_connectedContinuity_of_surjectivity_and_extensionData` show that it is enough for the monograph-level proof to supply rational-image coverage (or the stronger coordinate surjectivity) together with one-step extension data; the further bridges `rationalImage_and_ex` and `surjectivity_and_extensionData_of_refinementBisection_connectedContinuity_continuity` show that those sharper inputs themselves follow from the named rational-refinement/bisection output, the connected-continuity one-step bracket, restricted solvability, and, for surjectivity, full-coordinate continuity. The remaining raw construction work is exactly to derive the named bisection, connected-continuity bracketing, and continuity outputs from Wakker's bare restricted-solvability, connectedness, continuity, and Archimedean arguments, not to postulate calibrated integer refinement separately.

6 Roadmap for the remaining four certificates

Certificate	Current Lean consumer	Open content	Status
1 (hConstruct)	<code>pairwise_additivity_wakker_IV_2_7</code>	Standard-sequence construction (Wakker IV.2 Steps 1–4)	Multi-month, named input bundle in place
2 (hgloba1)	<code>global_additive_from</code>	Single-row coordinate pivot + chain construction	Forward chain proved; reverse + strict-monotonicity sub-residuals open
3 (haff)	<code>additive_rep_unique</code>	Diagnostic predicate retired; corrected predicate factors through bracketing + j-difference realization; bracketing-forall further factors through coordinate-utility unboundedness; on-coordinate consistency factors through triangle or the stronger coordinate-affine lift on j	This paper: auxiliary residuals, certificate decomposition, diagnostic refutation, corrected predicate, two-level further factoring, sample witness all theorem-backed; direct affine-lift route now follows from strict standard-sequence arithmetic plus structural monotonicity, rational-image coverage, and grid density; rational-image now reduces to continuity/interval solvability plus calibrated two-sided standard-sequence integer refinement; the universal all-standard-sequences grid-density target and the single selected strict dense-grid target are both formally refuted; the corrected rational-indexed utility mesh-family target is formulated and proved from rational-image coverage plus raw extension data; the raw Wakker seam is split into rational refinement/bisection and connected-continuity one-step bracketing certificates, with theorem-backed discharge from continuity plus calibrated integer refinement, theorem-backed converse assembly of integer refinement plus full-coordinate continuity from the raw-output seam and structural monotonicity, and theorem-backed reduction of both raw-output certificates to rational-image/surjectivity plus extension data.

7 Mechanization metrics

The complete artifact’s status as of this draft:

- Files: `WakkerInfrastructure.lean` (42 KB, sorry-free), `WakkerExistence.lean` (13 KB, sorry-free), `WakkerDebreuKoopmans.lean` (589 KB, sorry-free).
- Public theorem axiom audit: every public theorem depends only on `[propext, Classical.choice, Quot.sound]`. No rogue axioms anywhere in the artifact.
- Certificate roadmap table: `WakkerRoadmap.CertificateChecklist.explicitCertificateChecklist` mirrors the five-certificate layout of §4, with each certificate’s current status (`openTarget` / `consumerReady` / `discharged`).
- Companion artifact: the Management Science 2026 supplementary file `ClassicalLotteryInAction.lean` imports the W/DK consumer interfaces; nothing in the present paper’s discharge changes those interfaces, so the companion artifact remains stable.

8 Submission positioning

The natural venues for this paper are formal-methods journals and conferences whose review criteria emphasize formalization architecture, audit honesty, and reusable infrastructure rather than novel mathematical content. Three candidates:

- *Journal of Automated Reasoning* (JAR). Long-form submissions about Lean/Coq/Isabelle formalizations of substantive mathematical content. This paper’s certificate decomposition + Phase-0 necessity layer + M2 partial discharge is exactly the substance JAR has accepted in prior measurement-theory and decision-theory submissions (e.g. formalizations of Tarski’s axioms, of probability foundations, of game-theoretic equilibria).
- *Certified Programs and Proofs* (CPP). Conference-length submissions about formalization techniques. The paper-cum-artifact pair would target the formalization-architecture session.
- *Interactive Theorem Proving* (ITP). Similar profile to CPP; appropriate if the certificate-decomposition methodology is the headline contribution.

We recommend JAR as the primary target because the paper’s content fits a long-form journal review better than a conference page limit. The certificate decomposition methodology, the necessity-layer audits, and the M2 partial discharge each warrant several pages of explanation that conference venues cannot accommodate.

9 Boundary with the companion submission

This artifact is multi-month work whose remaining open content (Certificates 1, 2, 4, 5 plus the residual hexagon-condition obligation on Certificate 3) would not survive a single-session attempt. It does not block the Classical Lottery in Action submission to Management Science. The companion artifact cites this paper, at most, as ongoing work in preparation; the Management Science submission’s formalization status paragraph mentions only that the Wakker / Debreu–Koopmans representation layer is imported as named hypotheses, without claiming it is fully discharged.

The certificate roadmap in `WakkerRoadmap.CertificateChecklist` keeps the artifact’s open content auditable at every snapshot, so reviewers can verify that no claim of full discharge is being smuggled in.

10 Immediate next steps

- S1.** Done (Phase 0): close the necessity layer in `WakkerInfrastructure.lean §11` and the structural-axioms-only grid utility in `WakkerExistence.lean §1`, and mirror both in `WakkerRoadmap.CertificateChecklist`.
- S2.** Done: create the Lean certificate checklist with Prop-valued certificate shapes and the table `explicitCertificateChecklist`.
- S3.** Done: prove the first-layer projection lemmas in `WakkerRoadmap.CertificateChecklist` for the Wakker construction certificate.
- S4.** Done: factor the Debreu–Koopmans certificates analogously to the Wakker construction certificate.
- S5.** Done: preserve the Wakker wrapper theorems as regression tests.
- S6.** Done: discharge the M2 auxiliary residuals `hne` and `hpos` from coordinate essentiality, reducing the real-coordinate Certificate 3 frontier to the single `TradeoffTransferCertificate` hypothesis. See §5 of this paper.
- S7.** Done: factor `TradeoffTransferCertificate` into off-diagonal + on-diagonal pieces, fully discharge both pieces from a single named bridge (`UtilityValueRealizingEquivalence`) plus a triangle construction through a third coordinate. Six new sorry-free theorems pass the standard axiom audit. See §5.
- S8.** Done: refute the diagnostic `UtilityValueRealizingEquivalence_diagnostic` from any non-trivial additive representation; rename to `_diagnostic` and replace with a corrected predicate that lets the reference pair (v', w') vary; re-prove the off-diagonal discharge through the corrected predicate; further factor the corrected predicate into `JDifferenceRealizationCertificate` + `TradeoffBracketingForallCertificate` + `RestrictedSolvability`, theorem-backed by `utilityValueRealizingEquivalence_corrected_of_jDifferenceRealization_and_bracketing`.
- S9.** Done: discharge `TradeoffBracketingForallCertificate` from `CoordinateUtilityUnboundedCertificate` on each coordinate via the algebraic reduction of bracket inequalities under additive representation; sample witness for the canonical Bool/Real additive sum order proves the predicate is non-vacuous. Four new sorry-free theorems pass the standard axiom audit. In the real-coordinate case, theorem-backed cross-flow now further shows that interval-hitting image coverage—and hence `CoordinateRationalImageCertificate`—also closes this bracketing residual.
- S10.** Done: on real-coordinate domains, the stronger certificate `CoordinateSurjectivityCertificate` now discharges both `JDifferenceRealizationCertificate` and `TradeoffBracketingForallCertificate`, hence the corrected predicate `UtilityValueRealizingEquivalence`. Three new sorry-free theorems pass the standard axiom audit.
- S11.** Done: discharge the stronger `CoordinateSurjectivityCertificate` from continuity plus `CoordinateUtilityUnboundedCertificate` by the intermediate value theorem, and specialize this to structural monotonicity plus `CoordinateRationalImageCertificate`. The theorem

`utilityValueRealizingEquivalence_corrected_of_singleCoordinateMonotonicityAxiom_rationalImage` now closes the corrected M2 predicate from restricted solvability, structural monotonicity, and rational-image coverage.

- S12.** Done: reduce rational-image coverage to the next honest construction inputs. `CoordinateStandardSequenceIntegerRefinement` packages calibrated positive and reverse standard-sequence grids; `CoordinateIntegerSeedCertificate` isolates the exact 0, 1, -1 seed data needed to start those grids; `coordinateStandardSequenceIntegerRefinement` plus the integer- and rational-image corollaries, removes the strong coordinate-surjectivity hypothesis from this bridge layer; `coordinateIntegerImageCertificate_of_standardSequenceIntegerRefinement` turns calibrated grids into integer-image coverage; `coordinateUtilitySolvabilityCertificate_of_coordinateIntegerImageCertificate_of_standardSequenceIntegerRefinement` and `coordinateUtilitySolvabilityCertificate_of_continuity_univ` supply interval solvability; and `coordinateRationalImageCertificate_of_continuity_standardSequenceIntegerRefinement` realizes every rational by bracketing q between $\lfloor q \rfloor$ and $\lfloor q \rfloor + 1$.
- S13.** Done: split the raw Wakker refinement seam into two named construction outputs and prove their feed-through. `CoordinateRationalRefinementBisectionCertificate` gives rational-image coverage and `CoordinateIntegerSeedCertificate`; `CoordinateConnectedContinuityOneStepBracketCertificate` together with `RestrictedSolvability`, proves `CoordinateStandardSequenceExtensionData`; and `nonSurjectiveIntegerAndMeshBridgeOutputs_of_refinementBisection_and_connectedContinuity` packages the resulting integer-refinement and refined mesh-family outputs.
- S14.** Done: discharge both raw-output certificates from the current continuity plus calibrated integer-refinement machinery. `coordinateConnectedContinuityOneStepBracketCertificate_of_coordinateUtilitySolvability` proves the bracket by additive-representation algebra; `coordinateConnectedContinuityOneStepBracketCertificate` obtains its unboundedness input through rational-image coverage; `coordinateRationalRefinementBisection` packages rational hits as strict standard-sequence grid points; and `rationalRefinementBisection_and_connectedContinuity` bundles the two outputs.
- S15.** Done: prove the converse lower-target assembly from the raw-output refinement seam. `coordinateUtilityCombinatorics` turns rational refinement/bisection plus coordinate monotonicity into full-coordinate continuity; `integerRefinement_and_fullContinuity_of_refinementBisection_connectedContinuity_monotonicity` bundles continuity with calibrated integer standard-sequence refinement; and `integerRefinement_and_fullContinuity` replaces monotonicity by the raw structural `SingleCoordinateMonotonicityAxiom`.
- S16.** Done: reduce both raw-output certificates to sharper currently expressible construction inputs. `coordinateConnectedContinuityOneStepBracketCertificate_of_rationalRefinementBisectionCertificate` shows rational refinement/bisection already supplies the one-step bracket through rational-image unboundedness; `rationalRefinementBisection_and_connectedContinuity_of_rationalImage_and_unboundedness` proves both raw outputs from rational-image coverage plus one-step extension data; and `rationalRefinementBisection_and_connectedContinuity_of_surjectivity_and_extensionData` records the stronger surjectivity variant.
- S17.** Done: bundle the sharper inputs back from the named Wakker raw outputs. `rationalImage_and_extensionData` proves rational-image coverage plus one-step extension data from rational refinement/bisection, connected-continuity bracketing, and restricted solvability; `surjectivity_and_extensionData_of_refinementBisection` upgrades this to coordinate surjectivity plus extension data when full-coordinate continuity is available.
- S18.** Done: prove the conditional direct same-coordinate route `coordinateAffineLiftCertificate_of_commonStandardSequenceGridCalibration` shared normalized standard-sequence grid calibration for $R_1.V_j$ and $R_2.V_j$, plus continuity and

grid density, yields `CoordinateAffineLiftCertificate`, hence `OnCoordinateRatioConsistency`; also expose the `PairwiseGridNormalizationWitness` corollary.

- S19.** Done: prove the construction-side common-calibration route `coordinateAffineLiftCertificate_of_strict`. Along any strict standard sequence, both additive representations are affine arithmetic progressions on the same $\sigma.\alpha$ -grid, with positive step ratio; continuity plus grid density extends the affine relation globally.
- S20.** Done: prove the remaining input wrappers for that direct route. `coordinateAffineLiftContinuityInputs_o` supplies continuity of $R_1.V_j$ and $R_2.V_j$ from monotonicity plus rational-image coverage; `standardSequenceGrid` supplies density of the $\sigma.\alpha$ -grid from interval-hitting. The combined theorem `coordinateAffineLiftCertificate` closes the direct affine-lift route from exactly these construction-facing certificates.
- S21.** Done: prove the primitive-certificate stronger-output layer. `coordinateMonotonicityCertificate_of_singl` derives coordinate monotonicity from the raw structural monotonicity axiom; `coordinateSurjectivityCerti` derives coordinate surjectivity from structural monotonicity plus rational-image coverage; `standardSequenceG` derives interval-hitting from grid density; and `coordinateAffineLiftCertificate_of_strictStandardSequ` packages the direct affine-lift route through these sharper construction outputs.
- S22.** Done: refute both over-strong single-sequence grid-density targets. The theorem `additiveRealBool_not_star` (and its `false`-coordinate twin) uses the valid one-sided standard sequence $\sigma.\alpha(n) = n$ to refute the universal all-standard-sequences target. The stronger theorem `additiveRealBool_not_selectedRefined` (and its twin) shows that even one chosen strict standard sequence cannot be dense in the additive-real model, because strict standard sequences are arithmetic progressions.
- S23.** Done: formulate and prove the corrected refined mesh-family certificate. `CoordinateUtilityRefinedMeshFam` uses a rational-indexed family of strict standard sequences; `exists_strictStandardSequence_hitting_rati` embeds each rational target as a grid point of a strict sequence; `coordinateUtilityRefinedMeshFamilyCerti` proves interval-density of the family; and the family implies `CoordinateBetweenPointsCoverageCertificate` and `CoordinateDenseRangeCertificate`. A single dense standard sequence remains formally ruled out.
- S24.** Open: derive the named raw outputs `CoordinateRationalRefinementBisectionCertificate`, `CoordinateConnectedContinuityOneStepBracketCertificate`, and the full-coordinate continuity certificate from the bare monograph-level Wakker standard-sequence construction stack (restricted solvability, Archimedean, connectedness, continuity, and refinement/bisection). The new bridge theorems then yield rational-image coverage or coordinate surjectivity plus extension data automatically. Multi-week target.
- S25.** Open: discharge `hConstruct`. Multi-month target.
- S26.** Open: discharge `hglocal` via the chain-construction reverse direction. Multi-week target conditional on `hConstruct`.
- S27.** Open: discharge `hConc` via `QuasiToConcaveStrengtheningCertificate`. Multi-week target.
- S28.** Open: discharge `hPairConc` via the per-coordinate concavity transfer chain. Multi-week target conditional on Certificates 1, 4.